TOPIC 3 · INTERACTIVE DASHBOARDS WITH CUSTOM SKILLS
LAB 7 THEME · CUSTOM DASHBOARD SKILL

# Lab 7 — Create and Apply a Custom Interactive Dashboard Skill

| | |
|---|---|
| **Maps to** | LO3: build and validate an interactive Excel maintenance dashboard |
| **Duration** | 35 minutes |
| **Tools** | Excel Copilot · Microsoft 365 Copilot web · OneDrive |
| **Mock data** | data/Northstar_Work_Orders.xlsx |
| **Lab files** | data/templates/water-service-dashboard/SKILL.md (Reviewed interactive dashboard Skill fallback) |
| **Course** | GenAI for Engineers · v2.3 |
| **Prompts** | https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-7 |

## Goal

Create water-service-dashboard/SKILL.md for the Northstar maintenance workflow, install it in Excel, then invoke it to plan and build an interactive dashboard. Six KPIs and three charts share four filter selections. Workbook .Rules supplies scope, display and review constraints; the Skill supplies the repeatable method.

## What You Will Do

Create water-service-dashboard/SKILL.md for the Northstar maintenance workflow, install it in Excel, then invoke it to plan and build an interactive dashboard. Six KPIs and three charts share four filter selections. Workbook .Rules supplies scope, display and review constraints; the Skill supplies the repeatable method.

## What You Will Build

Northstar_Dashboard.xlsx with a Dashboard layout, six verified measures, three labelled charts and a measure dictionary ready for Lab 8 filters.

## Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- The clean table from earlier labs.

**Data note**

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

## Steps

**Step 1.** Open the synthetic source and data/templates/water-service-dashboard/SKILL.md. In Microsoft 365 Copilot on the web draft the full reusable Skill from this specification. Save it inside a matching water-service-dashboard folder and review every guard/measure before installation. Log the web prompt manually.

**PROMPT / CHECK — copy and adapt**

```
MICROSOFT 365 COPILOT WEB:
Write complete water-service-dashboard SKILL.md with YAML name/description. Require actual tblWorkOrders and report date in Control!B2; read visible .Rules. Preserve source IDs/statuses/owners, Technicians, Control, .Rules, Prompt_Log, Pareto_* and unrelated sheets. Plan first and ask before replacing only named dashboard outputs; rerun without duplicated charts. Dashboard B2:B5 select Site, Job_Category, Status, Assigned_To with All bypasses and AND logic. B7 total, B8 backlog excluding Closed/Cancelled, B9 backlog Due_Date before Control!B2, B10 Closed, B11 on-time Closed_Date<=Due_Date among Closed only with N/A if none, B12 recorded Cost_SGD across selected statuses. Follow visible .Rules DASHBOARD LAYOUT and STYLE: title A1; six KPI tiles in D4:I7 with Checks: PASS; three formula-backed charts (status column, category bar, full-width stacked site bar) with finding-led titles linked to Dashboard_Summaries; slicers for Site, Job_Category, Status and Assigned_To that filter every tile and chart through SUBTOTAL visibility. Use structured references, not fixed 120-row ranges, and never merge cells. Baseline check at 2026-09-30 All:120 total,48 Closed,12 Cancelled,60 backlog,50 overdue,30/48 on-time,35827.50 cost. Verify site/no-Closed/no-row tests, source preservation and a separate two-row append sandbox with 122 total,49 Closed,61 backlog,51 overdue,31/49 on-time and 36227.50 cost. These are snapshot tests, not universal future figures. Ask for missing inputs; never approve assignments, infer causes/savings or send email. Return the entire file.
```

**Step 2.** Use Excel Settings (...) > Manage plugins & skills > Custom skills > Open skills folder (Create OneDrive folder if needed). Copy the reviewed water-service-dashboard folder into that app-created location, Refresh, enable and verify its exact name in Choose plugins & skills. If unavailable, paste the entire reviewed Skill before each prompt, remove @water-service-dashboard and record Manual practice - native Skill unavailable. Installing maintenance-reporting or water-service-pareto does not install this distinct Skill.

**Step 3.** Open data/templates/Northstar_Dashboard_Starter.xlsx and save as Northstar_Dashboard.xlsx in OneDrive. If continuing your own workbook, copy its CLEAN tblWorkOrders into the starter, keeping exact headers and table name; do not create a second table. The starter has a labelled blank Dashboard and four selection cells B2:B5, initially All.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - CHAT ONLY:
@water-service-dashboard Inspect this Northstar Water Services dashboard starter. Confirm tblWorkOrders, 120 unique IDs, reporting date 2026-09-30 and selection cells B2:B5 initially All. Explain which source and control sheets must remain intact. Report discrepancies before any edits.
```

**Step 4.** Read Control and confirm the fixed reporting date. Ask for a visual plan before edits. Keep source, Checks and Prompt_Log separate from the Dashboard view. A dashboard is a decision interface: each visual should answer a maintenance question rather than decorate the page.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - PLAN:
@water-service-dashboard Plan Dashboard for Northstar maintenance at 2026-09-30. Use selection cells B2:B5 for Site, Job_Category, Status, Assigned_To, all initially All. Reserve B7 total orders, B8 backlog, B9 overdue backlog, B10 closed, B11 on-time closure rate, B12 Cost_SGD. Place three charts for status, category and site with labels, units and scope. Preserve source data.
```

**Step 5.** Ask Copilot to insert formulas for the six measures using structured table references. For now all selectors are All; in Lab 8 the same formulas must honour the four filters. Verify the denominator and use N/A, not 0%, if no Closed rows remain.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - ALLOW EDITING:
@water-service-dashboard Build the planned six KPI formulas from tblWorkOrders. Backlog excludes Closed/Cancelled. Overdue also requires Due_Date < Control!B2. On-time rate counts Closed_Date <= Due_Date among Closed rows only, and shows N/A when none. Recorded cost sums Cost_SGD of all included statuses. Link to table columns, never a fixed 120-row range. Keep source data unchanged.
```

**Step 6.** Add formula-backed chart summaries for the six statuses, five categories and five sites. Use bar/column charts: category and site are nominal groups; a line chart would imply an order that is not present. Label count axes and avoid 3-D effects. Then polish the charts with the built-in @chart-design skill from Lab 5.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - ALLOW EDITING:
@water-service-dashboard Create three editable bar or column charts using formula-backed
summaries: work-order count by Status, by Job_Category and by Site. Each summary must
reconcile to the total selected orders. Use white background, navy titles, blue/teal accents,
readable labels and a visible reporting date. Do not use category counts as evidence of causes
or use colour alone for status.
```

**Step 7.** In Checks confirm sum of each chart's categories equals 120 with All filters. Compare six KPIs to Control and record the on-time numerator/denominator. Add Evidence_Log entries for each chart and metric. If Copilot has used hard-coded numbers, replace them with formulas before Lab 8.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - CHAT ONLY:
@water-service-dashboard Reconcile the Northstar Water Services dashboard with Control at All
selections. Compare all six KPIs and totals of the Status, Job_Category and Site charts. Flag
hard-coded values, fixed row ranges, inconsistent scopes and any interpretation that treats
job-category frequency as a cause. Do not edit.
```

**Step 8.** Create a measure dictionary on a new Measures sheet: metric, business definition, formula, source fields, date rule, scope and limitation. Ask Copilot to explain formulas, then read one overdue example and one closed example from the source table yourself. Save the baseline dashboard with all filters All.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - CHAT ONLY:
@water-service-dashboard Explain each Dashboard measure in plain English, including terminal
statuses, strict overdue date rule, closed-only SLA denominator and meaning of recorded cost.
Give one source-row example for overdue and one for on-time closure; cite Work_Order_ID. Do
not guess a root cause or financial saving.
```

## Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

**Copy-ready prompts:** https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-7

| # | Purpose | Prompt |
|---|---|---|
| 1 | Scope | State actual source, report date and selected scope before acting. |
| 2 | Verify | Explain the source/formula checks and preserve unknowns and original records. |

## Test It

| Check | Expected result |
|---|---|
| Baseline | Total 120; backlog 60; overdue 50; closed 48. |
| SLA and cost | 30/48 = 62.5%; S$35,827.50. |
| Chart reconciliation | Three count summaries each total 120; formulas reference tblWorkOrders. |
| Decision design | Measures dictionary and source/date visible; no unsupported cause claims. |

## Troubleshooting

| If this happens | Fix |
|---|---|
| Copilot cannot find the source | Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app. |
| A total disagrees with Control | Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing. |
| The named feature is missing | Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran. |

## Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

## Reflection

Which evidence proves the output is correct, and which decision still needs a person?